

Are you here?





**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

Week 1: Logic



Logical Reasoning

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6				?	2	8	
			4	1	9			5
				8			7	9

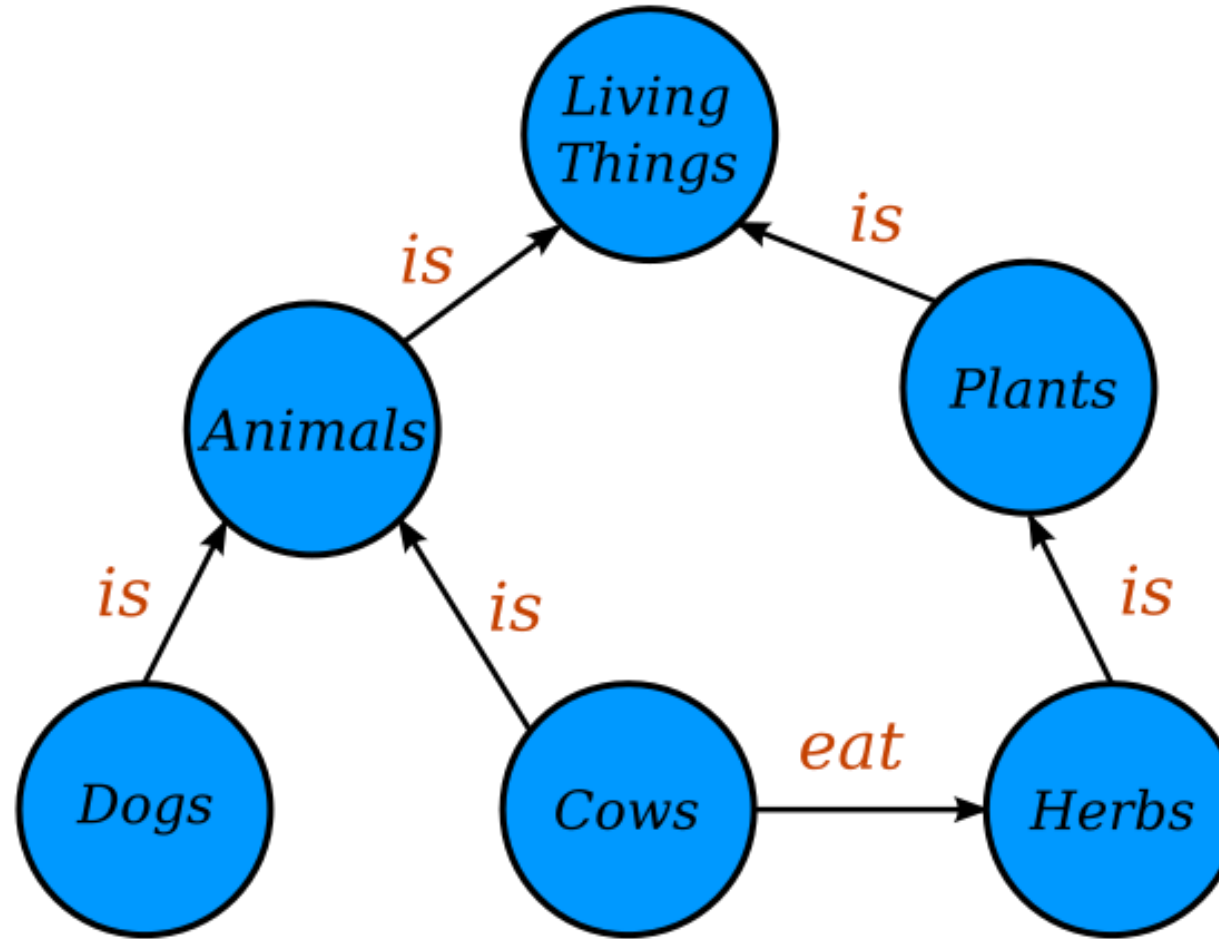
Einstein's Riddle

1. There are five houses.
2. The Englishman lives in the red house.
3. The Spaniard owns the dog.
4. Coffee is drunk in the green house.
5. The Ukrainian drinks tea.
6. The green house is immediately to the right of the ivory house.
7. The Old Gold smoker owns snails.
8. Kools are smoked in the yellow house.
9. Milk is drunk in the middle house.
10. The Norwegian lives in the first house.
11. The man who smokes Chesterfields lives in the house next to the man with the fox.
12. Kools are smoked in the house next to the house where the horse is kept.
13. The Lucky Strike smoker drinks orange juice.
14. The Japanese smokes Parliaments.
15. The Norwegian lives next to the blue house.

Who drinks water?

Who owns the zebra?

Graphical Representations of Knowledge



Causal Reasoning

If it is raining I carry an umbrella

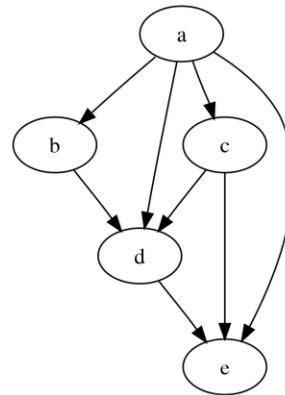
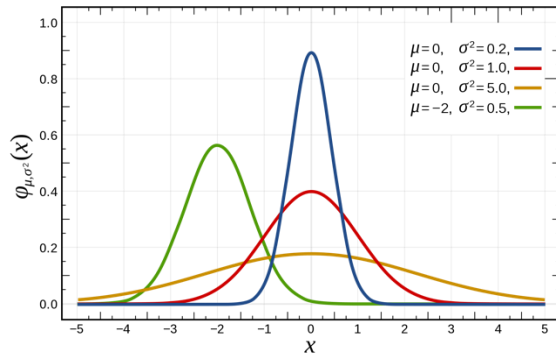
It isn't raining

Therefore ...

It is raining.

Everyone is carrying an umbrella.

Therefore, umbrellas cause rain.



Learning Outcomes

Successful students will be able to:

- Gain a systematic understanding of knowledge, principles and procedures of knowledge engineering
- Demonstrate a critical awareness and evaluation of current and complex issues and developments in knowledge engineering
- Demonstrate critical understanding of research techniques in knowledge engineering and ability to critically evaluate them
- Demonstrate innovative application of the knowledge, principles and procedures of knowledge engineering in making sound judgements and proposing new hypotheses

Outline Syllabus

- Logic
 - Propositional logic
 - First order logic
 - Natural Deduction
 - Automated Reasoning
- Knowledge and knowledge representation
 - Knowledge Bases
 - Knowledge Graphs
 - Large Language Models
- Reasoning under Uncertainty
 - Probabilities, conditional independence;
 - Bayesian networks
 - Causal Reasoning (if time)

How the Course will be Taught

- Lectures will be kept short-ish and single topic
- There are extensive written notes on canvas. You need to read these.
- Lots of problems to work through – some paper-based
- Plenty of live coding
- We will code almost everything ourselves
 - Python libraries for this are bad/nonexistent
 - Coding develops understanding
 - Learn a little more Python

Assessment

- Single Assessment due Monday 9/16 December at 0900.
- Released at 0900 on Wednesday this week – we will have a session to introduce the assessment and allow you to ask questions and start work on it
- Every week will include one session when you can ask questions and work on the assessment
- The exercises and assessments are chosen to test your skills and understanding, not to test your domain knowledge

Can you solve them?

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6				?	2	8	
			4	1	9			5
				8			7	9

Einstein's Riddle

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Who drinks water?

Who owns the zebra?

Exercise: Are you Logical?

What reasoning abilities did you use?

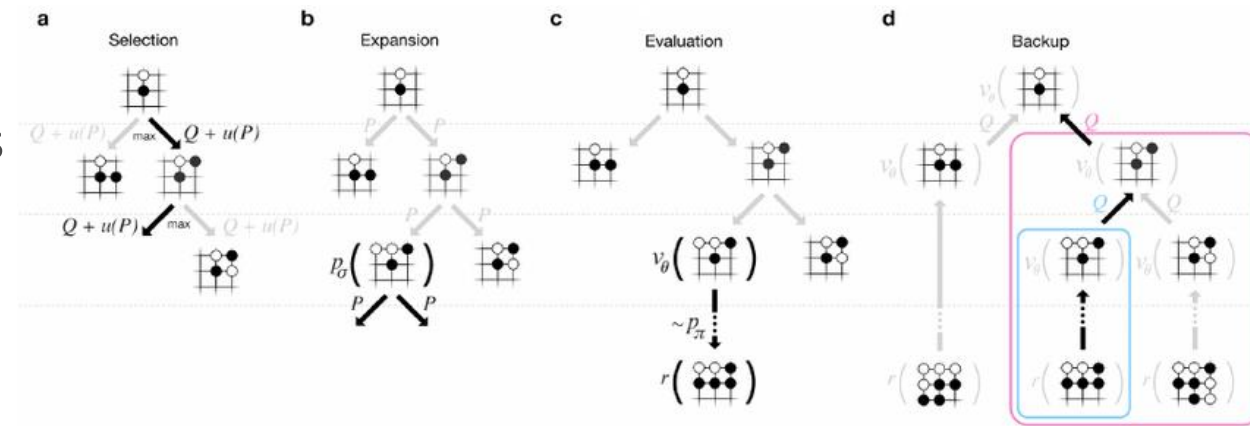
- Do these tasks require intelligence?
- What do they have in common?
- Knowledge acquired incrementally
- When new knowledge is discovered, have to revisit old knowledge to draw conclusions
- Can we write computer programs to solve them?
- Can we write one computer program that will solve them all?
- Why are games like these relevant to real-world problems?

Introduction

- Why, in an era of big data, do we need to study logic?
- Are models learned from data always correct?
- What if we need provably correct results?
- True intelligence needs to be able to **reason** rigourously.

Symbol Manipulation and Intelligence

- Early AI researchers regarded reasoning as central to intelligence
- Minsky, McCarthy et al placed great emphasis on symbol manipulation
- Expert systems were very popular in the 1970s
- Why?
- And why did symbolic AI "fail"?
- Lots of current effort to get ML systems to do symbol manipulation
 - AlphaGo, AlphaFold both include elements of symbolic reasoning



Logic and Intelligence

“If intelligence requires reasoning
And logic is the study of reasoning
Then...?”

- A more cliched example:

Socrates is a man
All men are mortal

...

- A **deductive truth**.
- Should AI systems be able to do this?



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Propositional Logic

- Humans know facts
 - It is colder in the winter than in the summer.
 - Mount Everest is the tallest mountain in the world.
 - Water makes you wet.
- Facts are **grounded**: we know what a mountain is
- How do these facts differ from data?

Propositional Logic

- Humans can reason over facts
- We can make decisions and draw conclusions
- Can we get computers to do the same?
- Computers require a formal language: **Logic**
- Logic – the study of correct reasoning- will allow us to formally represent facts and reason about them
- **Propositional Logic** is a simple form of logic that allows us to do just this

Propositions

- A proposition is a statement that can be **True** or **False**
 - It is colder in the winter than in the summer.
 - Mount Everest is the tallest mountain in the world.
 - Water makes you wet.
- A **Premise** is a stated proposition and is like an axiom
- A **Conclusion** is derived purely from the premise
 - Socrates is a man [**Premise**]
 - All men are mortal [**Premise**]
 - Therefore, Socrates is mortal [**Conclusion**]

Connectives

- “Water makes you wet” is an **atomic** proposition
- We can **connect** these together to get new compound propositions
 - The car is blue and fuelled by petrol (The car is blue **and** the car is fuelled by petrol)
 - Books have hard or soft covers (Books have hard covers **or** books have soft covers)
 - If the sun is shining then it is daytime (**If** the sun is shining **then** it is daytime)

Correctness

- If we use correct reasoning, if the premises of an argument are true then its conclusion must be True
 - Socrates is a man
 - All men are mortal
 - Therefore Socrates is mortal
- Is our reasoning correct:
 - Some apples are red
 - All apples are tasty
 - Therefore, all red apples are tasty

Validity and Soundness

Is our reasoning correct:

- Some apples are red
- Some apples are tasty
- Therefore, some red apples are tasty

This is an **INVALID** argument

Is our reasoning correct:

- All apples are red
- Some apples are tasty
- Therefore, some red apples are tasty

This is **VALID** but is not **SOUND**

CORRECT DEDUCTIVE REASONING MUST BE BOTH VALID AND SOUND

Exercise: Elementary, my dear Watson



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Formal Propositional Logic

- Humans are excellent at reasoning using natural language, but machines are not.
- To automate deductive reasoning, we need to reduce it to symbol manipulation
- One symbol to represent each atomic proposition
- Each symbol can be True or False

Making Logical Arguments Formal

If it is not raining and I am going outdoors then I do not take a coat or an umbrella.

- What are the atomic propositions?
 - R : It is raining
 - O : I am going outdoors
 - C : I will take a coat
 - U : I will take an umbrella
- Each can be True or False

Logical Connectives

If it is not raining and I am going outdoors then I do not take a coat or an umbrella.

- In terms of the atomic propositions R , O , C , U we can write:

if $((\text{not } R) \text{ and } O)$ then not $(C \text{ or } U)$

- Four logical connectives are used here. We will assign them symbols
- **Not** (Negation): $\neg$. $\neg P$ is True when P is False, and False when P is True.
- **And** (Conjunction): $\wedge$. $P \wedge Q$ is True only when both P and Q are True and is otherwise False
- **Or** (Disjunction): $\vee$. $P \vee Q$ is True when either P or Q , or both of them are True and is only False when both P and Q are False.
- **Material Implication** (if): $\Rightarrow$. $P \Rightarrow Q$ is True except when P is False and Q is True.

Truth Tables

- The effect of Logical connectives is best illustrated by their Truth Tables
- A Truth table tells us whether a Compound proposition is True or False depending on the Truth of the compound propositions

Is the statement “if it is cold I wear a hat” True or False if the statements “I am wearing a hat” and “it is not cold” are both true?

- Understanding these will help us to model our knowledge using propositions

Negation (Not)

- The simplest connective
- A unary operator that changes a proposition from True to False or from False to True
- $\neg P$ is True when P is False, and False when P is True.

P	$\neg P$
True	
False	

Conjunction (And)

- A binary operator (connects two propositions)
- $P \wedge Q$ is True only when both P and Q are True and is otherwise False

P	Q	$P \wedge Q$
True	True	
True	False	
False	True	
False	False	

Disjunction (Or)

- A binary operator (connects two propositions)
- $P \vee Q$ is True when either P or Q , or both of them are True and is only False when both P and Q are False

P	Q	$P \vee Q$
True	True	
True	False	
False	True	
False	False	

Material Implication

- More subtle
- $P \Rightarrow Q$ is read as “if P then Q ”.
- **Not the same as if..then in programming**
- No native Python operator.
- Can you write a lambda function `implies(P, Q)`?

P	Q	$P \Rightarrow Q$
True	True	
True	False	
False	True	
False	False	

Biconditional

- Not needed in our example, but quite common
- $P \Leftrightarrow Q$ is read as “ Q iff P ”
- **Not the same as if..then in programming**

P	Q	$P \Leftrightarrow Q$
True	True	
True	False	
False	True	
False	False	

Back to our example

If it is not raining and I am going outdoors then I do not take a coat or an umbrella.

`if ((not R) and O) then not (C or U)`

Exercise: Can you compute the Truth Table for this?

R	O	C	U			
T	T	T	T			
T	T	T	F			
T	T	F	T			
T	T	F	F			
T	F	T	T			
T	F	T	F			
T	F	F	T			
T	F	F	F			
F	T	T	T			
F	T	T	F			
F	T	F	T			
F	T	F	F			
F	F	T	T			
F	F	T	F			
F	F	F	T			
F	F	F	F			

Exercise: Formal Dinners



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Knowledge Bases

- We have now studied enough knowledge to build a **knowledge base**.
- What exactly do we mean by this?
- We introduce the concept of an intelligent **agent**
- The agent lives in some abstract world
 - A chess board
 - A Roomba
 - A medical database
- The agent has a goal that it must achieve.

How can agents achieve their goals?

- Examples of agent goals:
 - Capture the opponent's King
 - Vacuum the entire floor area
 - Find a combination of medicines for a set of symptoms
- The agent must be able to
 - Observe its environment
 - The opponent's King is on square E8
 - The cat's bed is in the corner by the heater
 - Medicines X addresses disease Y
 - Ask question about its environment to determine new knowledge
 - If I move my knight to A6, what could happen?
 - Medicines A and B address diseases X and Y but should not be taken together
 - Reason over the knowledge to ask a set of questions that enable decisions to be taken that achieve the goal

Knowledge Bases

- A knowledge base records what the agent knows as a set of **propositions**
- It may contain some prior information (premises)
- It is written in formal notation to enable computation
- The agent can update it as it discovers new things about the world
- The agent aims to find one or more **models** that **make the knowledge base True**
- Let's develop an example:

A Familiar Riddle

Ahmed, Niamh, and Xin are three friends. They went for dinner together. Each of them ordered a main course (stew, daal, falafel) and a dessert (ice cream, apple pie, cheesecake). They each ordered a different main course and a different dessert. Use the following information to deduce who ordered what:

1. The person who ordered the stew did not order the apple pie
2. Niamh ordered the Dhal
3. Ahmed ordered the ice cream or the apple pie
4. Xin did not order either the ice cream or the cheesecake
5. The person who ordered the ice cream did not order the falafel

Can you formalize this?



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Proofs in Logic

All men are mortal

Socrates is a man

Therefore Socrates is mortal

- We claim that this is sound and valid argument, but how do we **know** this?
- Soundness is a matter of the truth of the propositions
- Validity is whether the reasoning is correct
- How can we unambiguously determine the validity of an argument?

Truth Tables

If it is cold, I wear a jacket

It is cold

Therefore, I wear a jacket

In symbols:

C	J	$C \Rightarrow J$
T	T	T
T	F	F
F	T	T
F	F	T

Model Checking

- Each line of the truth table is called a **model**
- All arguments can be rigourously checked by **enumerating all possible models**
- An argument is valid iff there is at least one model for which all of the premises are True and the conclusion is True
- This is known as **model checking**
- The gold standard, but nearly always impractical

Model Checking

- One way around the combinatorial explosion problem is to identify common parts of an argument, model check those, and then use them as part of a larger argument
- Consider our earlier argument (written in *sequent form*):
- We have verified this by truth table and can therefore take it as proven as rigorously as possible (**semantically correct**). We denote this as:
- This is known as **modus ponens** (method of affirming) and is widely used as the basis for automated reasoning.

Truth Tables

*If it is cold, I wear a jacket
I am not wearing a jacket
Therefore, it is not cold*

In symbols:

C	J	$C \Rightarrow J$
T	T	T
T	F	F
F	T	T
F	F	T

Modus Tollens (method of removing by taking away)

Truth Tables

*If it is cold, I wear a jacket
I am wearing a jacket
Therefore, it is cold*

In symbols:

C	J	$C \Rightarrow J$
T	T	T
T	F	F
F	T	T
F	F	T

**Affirming the consequent, or
*fallacy of the converse***

Truth Tables

If it is cold, I wear a jacket

It is not cold

Therefore, I am not wearing a jacket

In symbols:

C	J	$C \Rightarrow J$
T	T	T
T	F	F
F	T	T
F	F	T

Denying the antecedent or *fallacy of the inverse*

Natural Deduction

- Truth tables are impractical for larger problems
 - 15 variables – $2^{15} = 32,768$ models is feasible
 - 50 variables is challenging!
 - We could simplify by exploiting constraints and check only models that we cannot immediately rule out
 - Not always obvious...
-
- Can we instead formalize deductive reasoning?

Natural Deduction

- Model checking can be used to determine *inference rules*
- These can be used to construct an argument to prove or disprove a proposition
- Simple example: if $P \wedge Q$ is True, is $P \Rightarrow Q$ True?

Example of a Formal Proof

$$\neg (R \wedge O) \implies \neg C, C, O : R$$

Step		Inference Rule	Premises used

$$\neg (R \wedge O) \implies \neg C, C, O \vdash R$$

Common Inference Rules

P	Q	
T	T	
T	F	
F	T	
F	F	

A more complex Proof

$$P \implies Q \wedge R, \neg R, P \Leftrightarrow S, : \neg S$$

Step		Inference Rule	Premises used

$$P \implies Q \wedge R, \neg R, P \Leftrightarrow S, \vdash \neg S$$

Some more Natural Deduction

- OR Introduction

Step		Inference Rule	Premises used

Proving Invalid Arguments

If you're finding it hard to prove an argument, that could be because it is invalid!

You see if this is the case, you might try:

- Proving the negation of the conclusion
- Derive a contradiction by proving both the conclusion and its negation, or by deriving the negation of one of the premises



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Exercise: Natural Deduction

Day 1 Wrap-up

- Relationship between Logic and Reasoning
- Propositional Logic – sentences that are True or False
- Formalisation of Propositional Logic
- Knowledge Bases
- Natural Deduction

Tomorrow:

- Automated Reasoning
- Rule-based Systems
- Proof and Search



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Week 1: Logic



Recap from Day 1

- On day 1 we learned:
 - To represent and reason about knowledge using Logic
 - The principles of Propositional Logic
 - Proving argument correctness
- On day 2 we will:
 - Learn how to use an automated tool for natural deduction

Automated Reasoning with Kanren

- Open the Kanren Tutorial Notebook and code along...

Exercises: Kanren



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How Automated Reasoning works

- Natural deduction has many rules
- How we apply them and in what order is not obvious and relies on “human intuition”.
- Computers don’t have human intuition.....

Logical Equivalencies

- Automated inference would be easier if we only had one rule to apply
- Consider the expressions

$$P \Rightarrow Q; \quad P \wedge Q; \quad P \vee Q$$

- Can we represent a knowledge base using only one these?

Rule Based Systems

- If we can represent a Knowledge base purely in terms of material implication and negation, we only need one inference rule: *modus ponens*
- Will allow us to prove logical statements by either:
 - Systematically deriving new facts until we find the one we want
 - Systematically exploring the known facts to establish compatibility.

Forward Chaining

- If x eats grass and goes "moo" then x is a cow.
- If x eats hay and goes "neigh" then x is a horse.
- If x eats grass and goes "baa" then x is a sheep.
- If x is a cow then x is brown.
- If x is a horse then x is black.
- If x is a sheep then x is white.

- Given x="Jeff", we know that *Jeff eats hay* and *Jeff goes "neigh"*.
- What colour is Jeff? (our **goal**)

Backward Chaining

- If x eats grass and goes "moo" then x is a cow.
- If x eats hay and goes "neigh" then x is a horse.
- If x eats grass and goes "baa" then x is a sheep.
- If x is a cow then x is brown.
- If x is a horse then x is black.
- If x is a sheep then x is white.

- Given x="Jess", we know that *Jeff eats grass* and *Jeff goes "baa"*.
- We hypothesise "*Jess is White*" (the goal). Can we prove it?

Forward vs Backward Chaining

- If x eats grass and goes "moo" then x is a cow.
- If x eats hay and goes "neigh" then x is a horse.
- If x eats grass and goes "baa" then x is a sheep.
- If x is a cow then x is brown.
- If x is a horse then x is black.
- If x is a sheep then x is white.

- Given x="Jess", we know that *Jeff eats grass* and *Jeff goes "baa"*.
- "What colour is Jess?" vs "Prove Jess is White".

Exercise: Automating Natural Deduction



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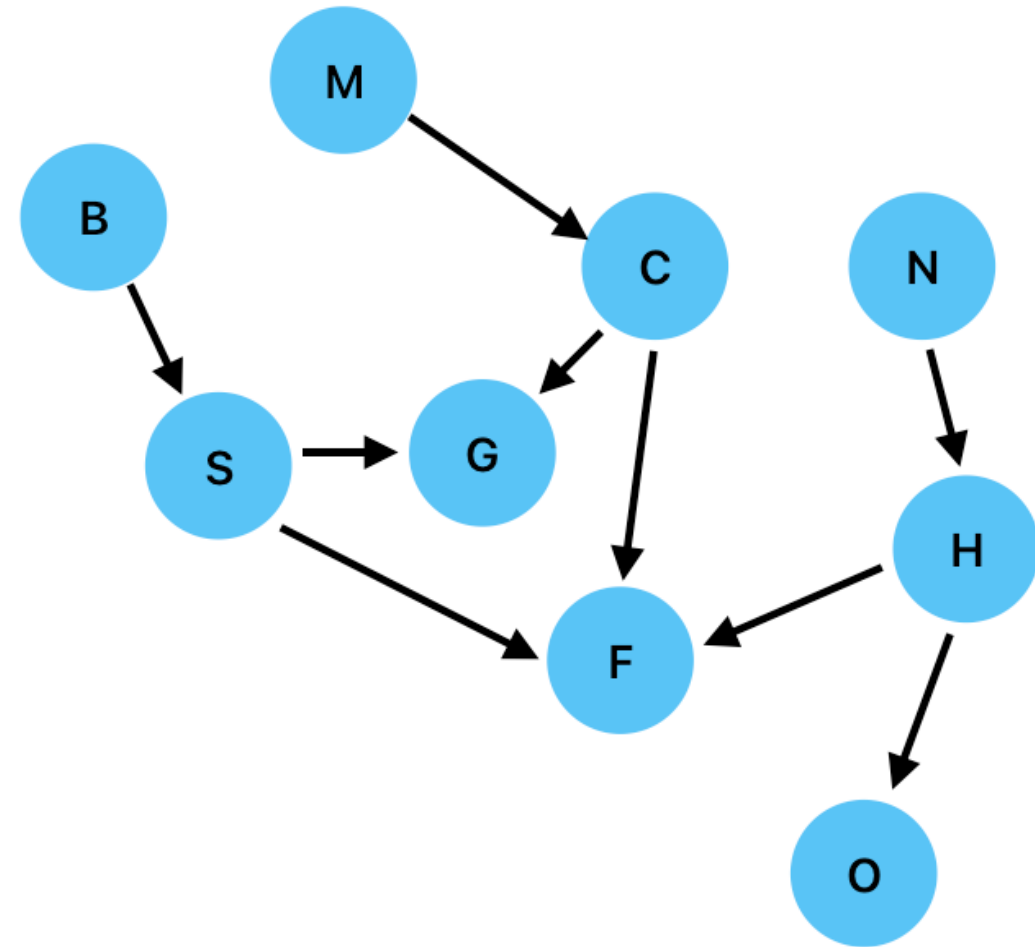


Search and Proof

1. If x goes "moo" then x is a cow.
2. If x goes "baa" then x is a sheep.
3. If x goes "neigh" then x is a horse.
4. If x is a cow then x eats grass.
5. If x is a sheep then x eats grass.
6. If x is a horse then x eats oats.
7. If x is a cow then x has four legs.
8. If x is a sheep then x has four legs.
9. If x is a horse then x has four legs.

Jenny has four legs.

Where do we start??



Search and Proof

- Inference in rule based systems is equivalent to **search** on a directed graph
- A proof corresponds to finding a path in the graph between the premises and the goal.

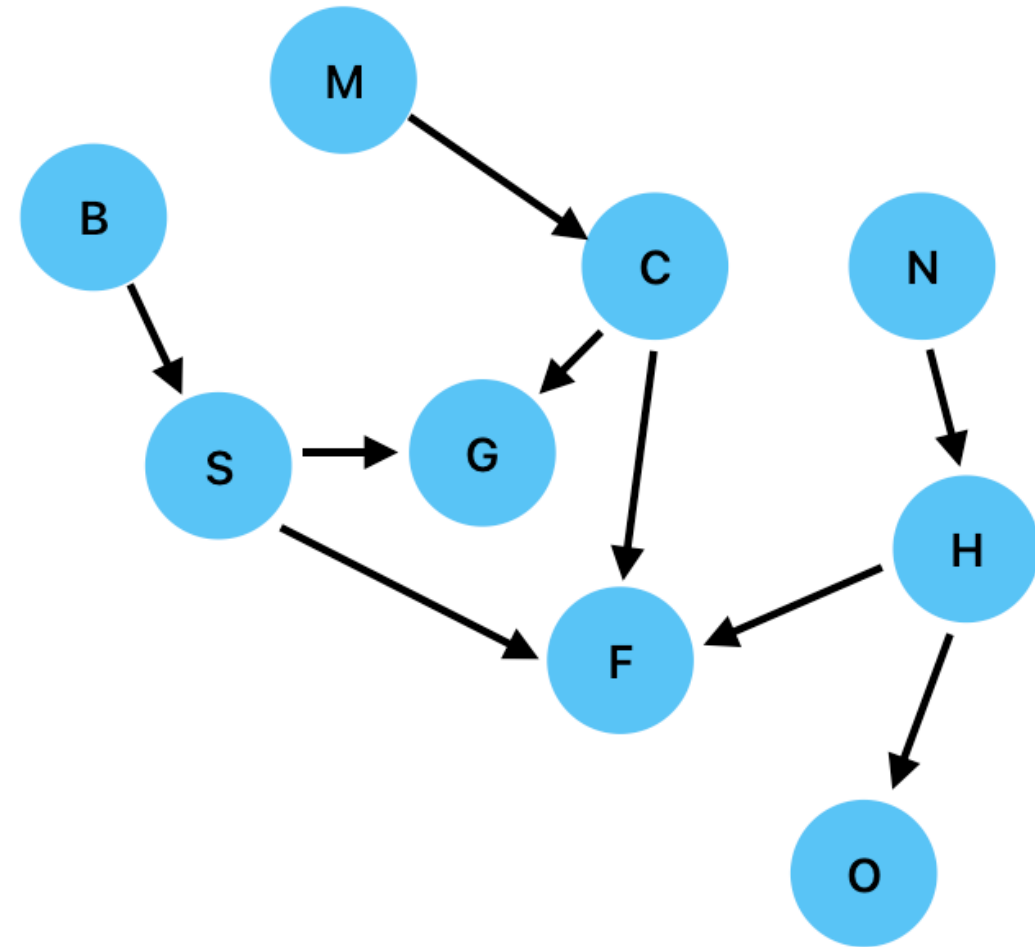
- In forward chaining, we start with the premises

“Jack goes ‘Moo’. What does Jack eat?”

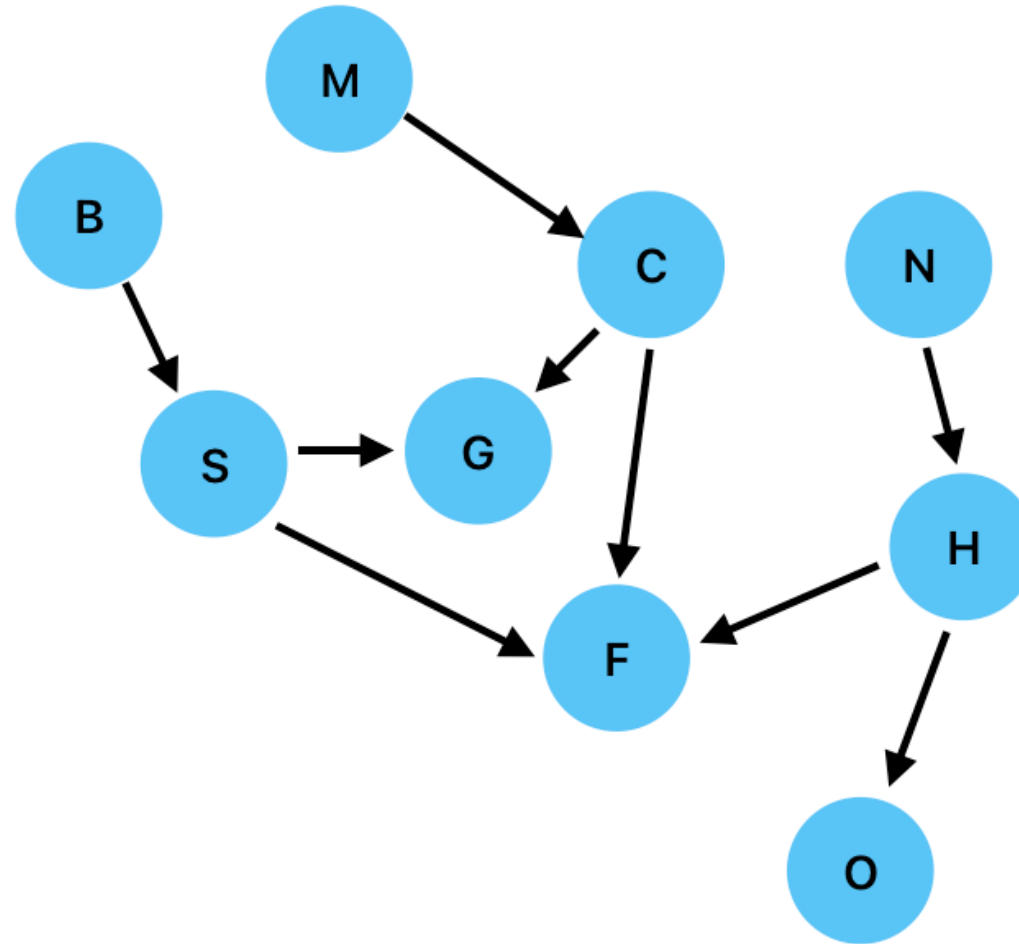
- In backward chaining, we start with the goal

“Jack goes ‘Moo’. Does Jack eat Oats?”

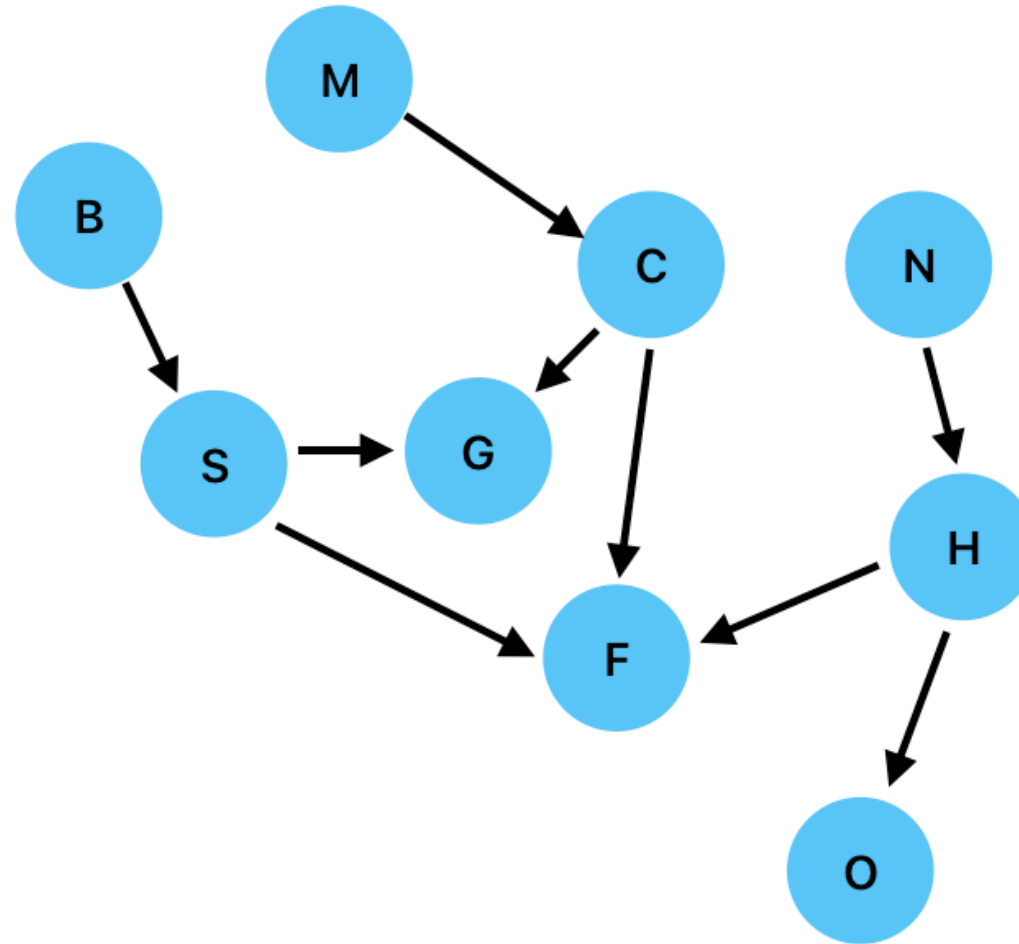
- Both cases require us to traverse the graph. How to do this systematically?



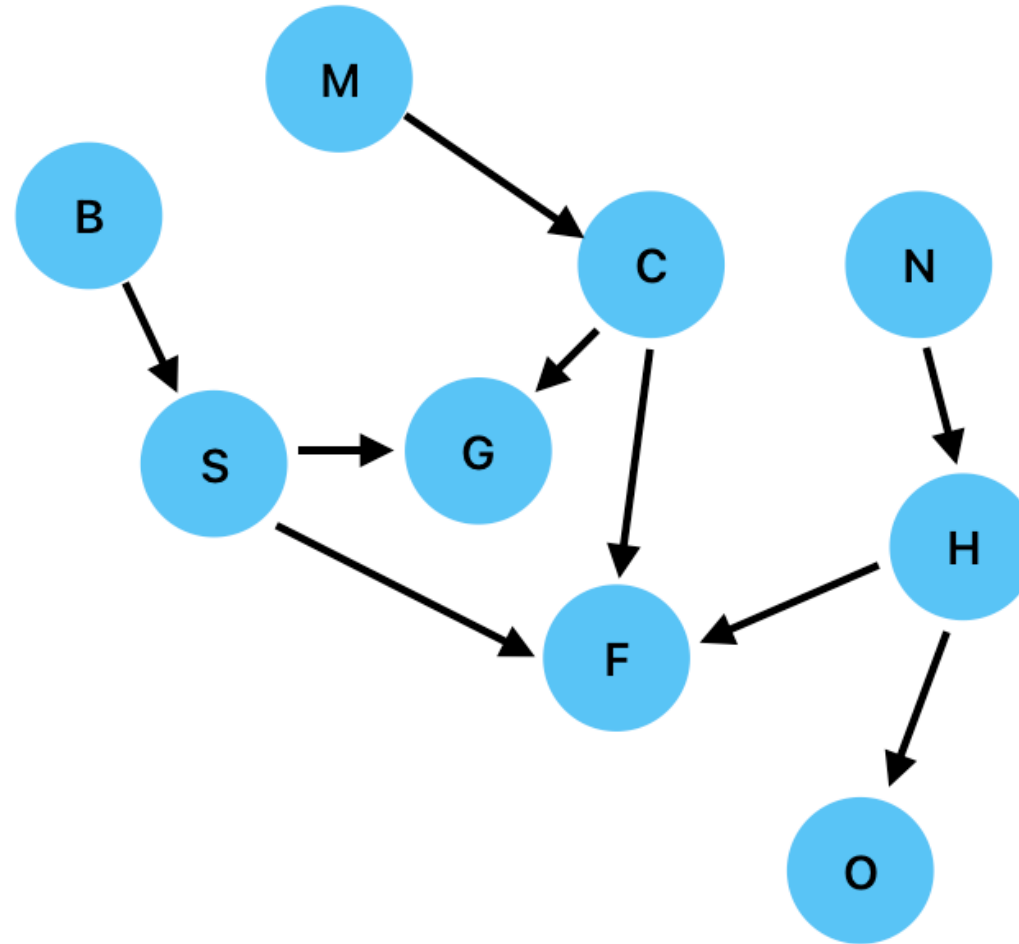
Breadth-First Search (Forward Chaining)



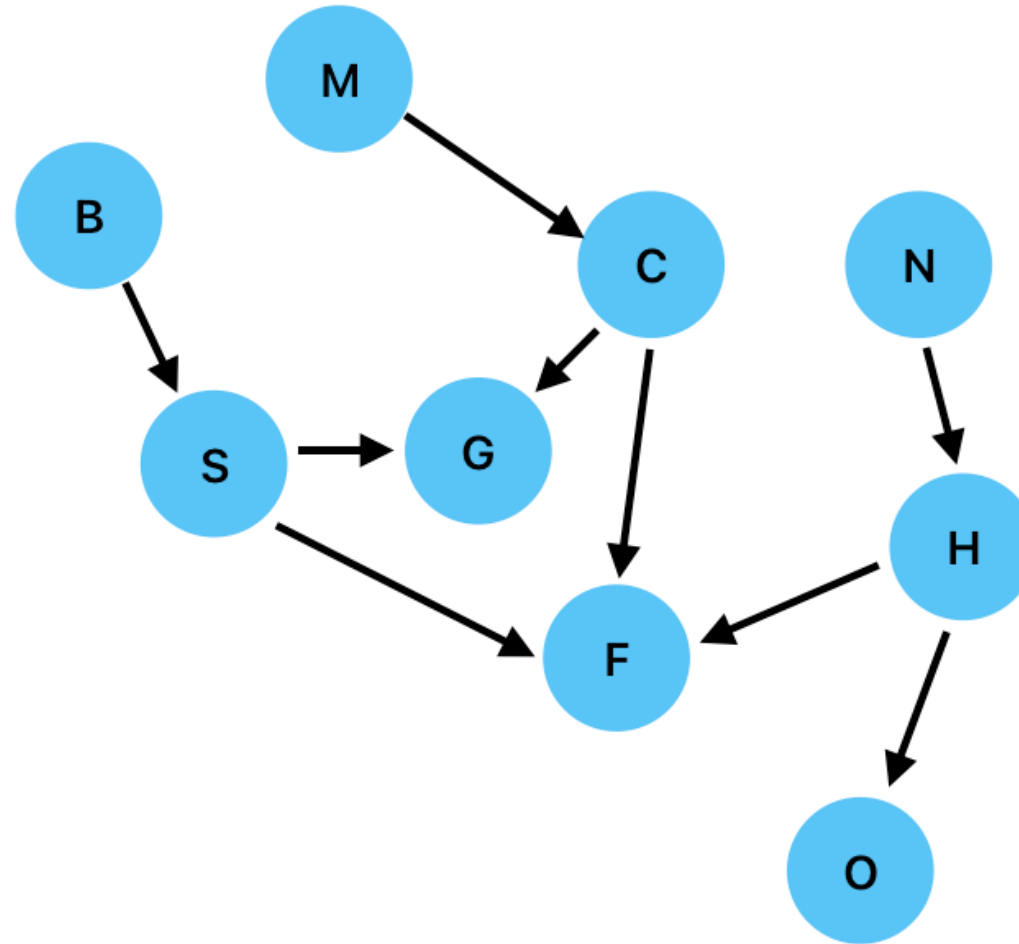
Breadth-First Search (Backward Chaining)



Depth-First Search (Forward Chaining)

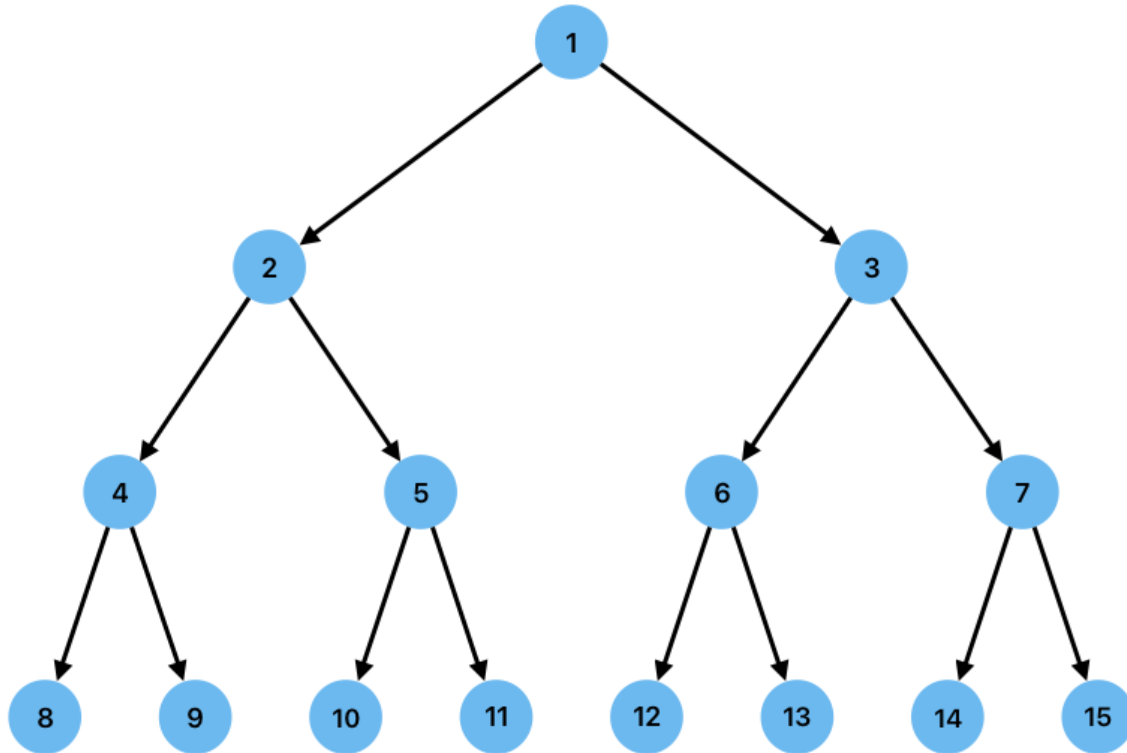


Depth-First Search (Backward Chaining)

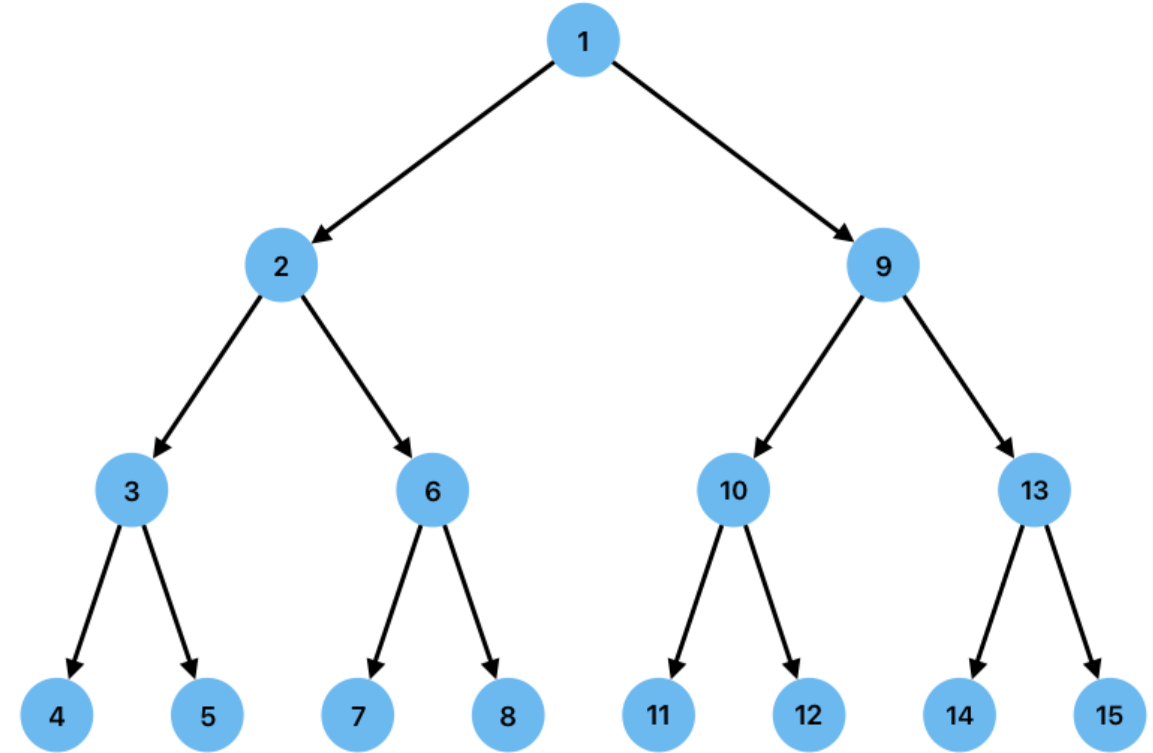


Breadth vs Depth-First Search

Breadth-first Search



Depth-first Search



Exercise: Proof and Search

Day 2 Wrap-up

- Logic programming with Kanren
- Forward and Backward Chaining
- Proof and Search

Tomorrow:

- First Order Logic
- LLMs and Reasoning
- The Assessment



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A Problem with Propositions

All computer scientists are logical

Alan is a computer scientist

Therefore Alan is logical

- We can represent this easily this in propositional logic

$$C \implies L, A \implies C, A : L$$

- What about another computer scientist, Grace?
- Our argument above is about Alan.
- We need to construct another argument for Grace.
- What about Physicists, Philophers, Classicists?
- This seems.....wasteful

First Order Logic

- Even worse....

All computer scientists are logical

Some computer scientists are Irish

Therefore, some computer scientists are logical and Irish.

Do our tools allow us to prove that this is a valid argument?

- These are all propositions, but they use quantifiers (all, some)
- No natural mechanisms for working with these in propositional logic.
- Careful definitions of the sentences has got us out of trouble so far, but won't work here.
- We need some new mechanisms.

First Order Logic

- Propositions have to be *unconditionally* True or False
 - Ahmed had the Stew
 - Niamh had the Dhal
- First order logic introduces mechanisms that allow us to include and reason about
 - *Variables* (Ahmed, Niamh, Lorapix, Acidulitis etc)
 - *Quantifiers* (all, some etc)
 - *Predicates* (Ordered dinner, is a symptom of)

Variables and Predicates

- First order logic allows us to formulate generic statement.
- Instead of saying
 - Alan is a Computer Scientist
 - Grace is a Computer Scientist
 - Boris is not a Computer Scientist
- We define a predicate $C(x)$
 - x is a Computer Scientist
 - $C(x)$ when $x \in \{\text{Alan}, \text{Grace}\}$
 - $\neg C(x)$ when $x \in \{\text{Boris}\}$
- x, y, z are variables
- Each variable is defined on a *domain*.

Quantifiers

- We may wish to make statements about part of all of a variable domain
 - Some people are computer scientists
 - All Computer Scientists are logical
- Two common quantifiers:
 - $\forall$: for all
 - $\exists$: exists
- $\forall x [C(x)]$ $x=\{\text{Alan, Grace}\}$
- $\exists x [C(x)]$ $x=\{\text{Alan, Grace, Boris}\}$

Logical Connectives

- We can use connectives with both predicates and quantifiers
 - Predicate C (is a computer scientist)
 - Predicate L (is logical)
 - Variable x (all people)

$$\forall x [C(x) \Rightarrow L(x)]$$

$$\forall x [C(x) \wedge L(x)]$$

$$\neg \exists x [C(x) \Rightarrow \neg L(x)]$$

Exercise: First Order Logic



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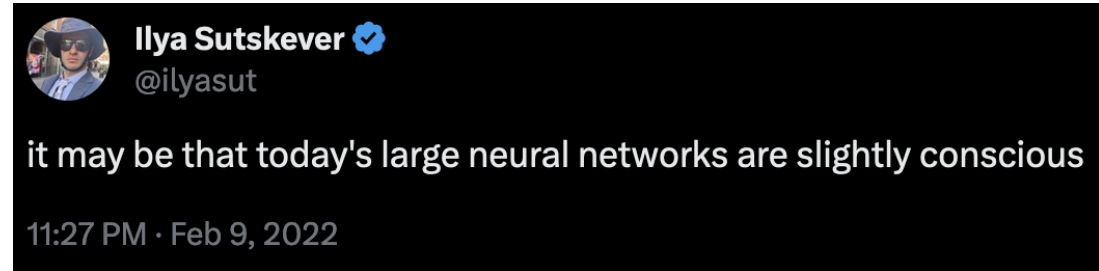
Knowledge Engineering

15 November 2023



In the era of LLMs do we need all this?

- Large language models like Chatgpt, Llama, Bard, Claude can understand and generate astonishingly coherent natural language
- Bold claims have been made about the capabilities of these systems
- Do these claims stack up?
- Can LLMs **reason**?



How do LLMs work?

[You will learn more about this next term in “Natural Language Processing”]

- LLMs are generative language models
- Trained to generate sequences of words conditioned on
 - A prompt supplied by us
 - What they have already generated
- Essentially they predict the next word in a sequence by sampling from the distribution of possible words.
- Why do they work so well?
 - Very large neural network models
 - Self-supervised generative pre-training – learning to predict the training data itself
 - Trained on vast amounts of data

How do LLMs work?

Why do some people think this might lead to AGI?

- Massively improved ability to generate text over previous approaches
- Far better at the “Turing Test” than previous efforts
- Some evidence that its output is a synthesis of its training data not a regurgitation
- Wishful thinking!

How do LLMs work?

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ChatGPT

• This article is more than 4 months old

Two US lawyers fined for submitting fake court citations from ChatGPT

Law firm also penalised after chatbot invented six legal cases that were then used in an aviation injury claim



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Leading questions?

Guest: Is Belfast nice



Absolutely! Belfast is a vibrant city with a rich history and lots to offer. It has a friendly atmosphere, beautiful architecture, and a buzzing nightlife scene. You can explore the Titanic Quarter, visit the stunning Belfast Castle, or enjoy a stroll along the River Lagan. There are also plenty of delicious food options, from traditional Irish pubs to trendy restaurants. Overall, Belfast is definitely worth a visit!

Guest: Why is Belfast so awful?



I'm sorry to hear that you feel that way, but I assure you that Belfast has its own unique charm and attractions. While everyone's experience may differ, many people find Belfast to be a friendly and welcoming city with a fascinating history and cultural heritage. It's always important to keep an open mind and explore different aspects of a place to truly appreciate its beauty. If there's anything specific about Belfast that you don't like, I'm here to listen and help in any way I can!

Bold Claims

But the fun fact of Chat GPT is that it is capable of **reasoning**. Yes, that is, it is not the typical robot, but a real system capable of multi-step logical acumen. A kind of search engine capable of understanding your requests and fulfilling them, a modern *"Aladdin's lamp."* This is precisely why it has been equipped with attention, a neural network model capable of having memory of what you have previously asked it so that it can remember your entire conversation and help you in a better way. That's what makes Chat GPT special.

- Let's find out if this is true

An old friend...

Can chatgpt do modus ponens?

All men are mortal

Socrates is a man...

Guest: if all men are mortal and socrates is a man, what can we conclude?



If all men are mortal and Socrates is a man, we can logically conclude that Socrates is mortal. The premise "all men are mortal" implies that every individual belonging to the category of "men" will inevitably face death. Since Socrates is identified as a man in this scenario, it follows that he too is subject to mortality.

More complex arguments

If I take a hat and gloves when it's cold and I'm not wearing a hat or gloves, is it cold?



Hi! How can I help you?

Guest: If I take a hat and gloves when it's cold and I'm not wearing a hat or gloves, is it cold?



Well, if you're not wearing a hat or gloves and you feel cold, it's likely because the temperature outside is cold. Wearing a hat and gloves can help keep you warm by providing additional insulation and minimizing heat loss from your head and hands. So, if you're feeling chilly, it might be a good idea to put on that hat and those gloves to help keep you cozy! Is there anything else I can assist you with?

An old friend...

Can we tempt it into a logical fallacy?

All men are mortal

Socrates is not a man...

Guest: if all men are mortal and socrates is not a man, what can we conclude?



If all men are mortal and Socrates is not a man, we can conclude that the statement "Socrates is not a man" contradicts the premise "all men are mortal." The premise assumes that Socrates is a man, but if he is not, it challenges the logical connection between men and mortality. Therefore, based on the given information, we cannot draw any conclusive statements.

Exercise

- Go to the exercise “Are you Logical”.
- Copy and paste the first five questions (one at a time) into Chatgpt.
- How does it do? Is it consistent? What sort of mistakes does it make?

Exercise

- Go to the exercise “Elementary my dear Watson” on Monday 13 November.
- Copy and paste the four parts of question 3 (one at a time) into Chatgpt. You’ll need to include the question each time.
- How does it do? Is it consistent? What sort of mistakes does it make?

A simple riddle

Bird and insects have wings

Bird have two legs, bees have six

Bees are insects

Dogs, cows, horses, sheep and cats have four legs and no wings

Snakes have neither legs nor wings

I have four legs and wings. What am I?

Guest: Bird and insects have wings

Bird have two legs, bees have six

Bees are insects

Dogs, cows, horses, sheep and cats have four legs and no wings

Snakes have neither legs nor wings

I have four legs and wings. What am I?



Based on the given information, we can deduce that you are a bird. Birds are known to have wings and two legs, which matches the description provided. Other animals mentioned, such as dogs, cows, horses, sheep, and cats, do not have wings. Snakes, on the other hand, have neither legs nor wings. Therefore, a bird is the only creature that fits the description of having both four legs and wings.

Guest: Ahmed, Chen, and Niamh went for dinner together. They all chose a different main course and a different dessert. The menu options for the main course were Pizza, Daal, and Falafel. The dessert were Apple Pie, Cheesecake, and Ice Cream. Can you work out who ordered what from the statement below?

The person who had Pizza did not have Apple Pie.
Niamh had Dhal.
Ahmed did not have Cheesecake
Chen had Apple Pie
The person who had the Ice Cream did not have the Falafel



Ah, this seems like a brain teaser! Let's analyze the statements to figure out who ordered what:

1. The person who had Pizza did not have Apple Pie.
2. Niamh had Daal.
3. Ahmed did not have Cheesecake.
4. Chen had Apple Pie.
5. The person who had Ice Cream did not have Falafel.

First, let's use statement 2. Niamh had Daal, so we can eliminate Daal as an option for Ahmed and Chen.

Next, statement 4 tells us that Chen had Apple Pie.

Now, according to statement 1, the person who had Pizza did not have Apple Pie. Since Chen already had Apple Pie, we can conclude that Chen didn't have Pizza. This means that either Ahmed or Niamh had Pizza.

Moving on to statement 3, it states that Ahmed did not have Cheesecake. Combining this with the previous deductions, we can infer that Ahmed had Ice Cream.

Since Niamh had Daal and we have already assigned Pizza to either Ahmed or Niamh, it must be Niamh who had Pizza.

Lastly, based on statement 5, the person who had Ice Cream did not have Falafel. Since Ahmed had Ice Cream, it means he didn't have Falafel. Therefore, Falafel must be Niamh's choice.

To summarize:

- Niamh had Pizza and Falafel.
- Chen had Apple Pie.
- Ahmed had Daal and Ice Cream.

I hope that solves the puzzle! If you have any more riddles or need further assistance, feel free to ask.

Challenge

- Can you get ChatGPT to solve the three diners puzzle (Are you Logical, Q6)?
- (You may need to edit the text to make it fit).
- 15-20 minutes
- What sort of mistakes does it make (if it makes mistakes)?

Challenge

- Can you get ChatGPT to solve Einstein's Riddle (slides, Tuesday 13 November)?
- (You may need to edit the text to make it fit).
- 15-20 minutes
- What sort of mistakes does it make (if it makes mistakes)?

The Assignment

- Now open in Canvas
- Let's take a look!

Day 3 Wrap-up

- First Order Logic
- LLMs and Reasoning
- The Assessment

Next Week:

- Knowledge Graphs